

Computational Study of the Structure and Transport in Pyrrolidinium-Li-TFSI-Silica Ionogels

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Cite This: *J. Phys. Chem. B* 2021, 125, 13003–13014



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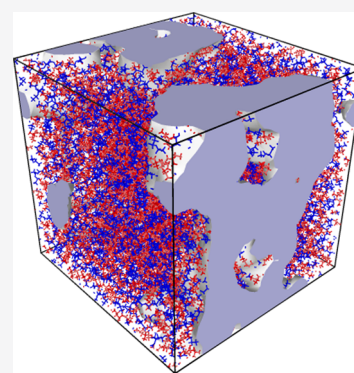


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ABSTRACT: Ionogels (IGs) are a unique class of composite materials with attributes that make them promising materials for applications in electrochemical energy storage. Due to the solid porous matrix that confines the ionic liquid (IL) in the IG, they can be used as self-supporting electrolytes. Furthermore, interactions of the IL with the porous matrix can have beneficial effects on transport, such as lowering the freezing/glass transition temperature of the conducting IL. In this work, we employ molecular dynamics simulations to investigate the influence of the porous morphology and solid volume fraction on ionic conductivity and Li^+ diffusivity using a representative 0.5 M Li-*bis*(trifluoromethane)sulfonimide (TFSI)-pyrrolidinium (Pyr1.3) IL confined in a nanoporous silica matrix. The effect of the morphology of the confining matrix is compared using the pure IL as a baseline. We find that the tracer and collective Li^+ diffusion and ionic conductivity of all the model IGs have significantly lower temperature dependence than the corresponding pure IL. In general, low-silica IGs with wide pores displayed the best transport properties at high temperatures, but the trends with the morphology for the nested set of transport coefficients we examined changed as the collective behavior of the Li^+ ions and the molecular IL components were considered. Remarkably, some of the model IGs displayed better transport properties on a volume of fluid basis at low temperatures than the constituent IL. These trends were tied to structural changes revealed by the radial distribution functions of the IL components and the silica surface, including a decreasing Li^+ adsorption peak of the surface silica indicating a change in the relative contributions of bulk-like and surface-like transport in the confined IL.



1. INTRODUCTION

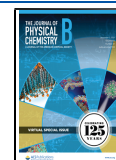
An ionogel (IG) is a conducting ionic liquid (IL) confined in a mesoporous or nanoporous solid matrix. IGs are promising candidates for a variety of electrochemical applications. By themselves, ILs have desirable electrochemical properties such as the following: high ionic conductivity, a wide electrochemical potential window, nonflammability, and nonvolatility.^{1,2} Confining an IL in a nanoporous matrix provides both sufficient mechanical strength for an IG to act as self-supporting electrolyte and confinement effects that benefit transport properties, such as depressing the glass transition temperature.³ The porous structure of the matrix is a physically controllable characteristic of the IG that influences its electrochemical performance and, therefore, can be optimized. There are a number of trade-offs in the design of an IG for selected IL and matrix materials. In particular, the balance between strength and conductivity in a silica IG is largely determined by the silica fraction. Increasing the solid fraction will strengthen the IG but comes at the cost of reducing ionic conductivity because the IL fraction is its main determining factor. There are also more subtle effects of morphology beyond the simple, first-order influence fluid volume fraction, but these have largely gone unstudied.^{4,5}

Overall, ILs and IGs have been subject to intense study recently as the need for electrochemical storage has increased. A number of new materials and material combinations, and experimental measurements have been produced. Extending the liquid range of ILs for battery operation had been a primary focus of many studies because the transition temperature tends to be near ambient conditions. For example, Kunze et al.⁶ lowered the melting point of pyrrolidinium (Pyr) ILs by tuning binary mixtures. Improving properties via nanoconfinement has been a theme with IG investigations. Göbel et al.⁷ created mesoporous silica with a sol–gel process and filled it with a variety of ILs using a methanol–IL exchange technique. They found that melting and glass transition of the IL was shifted to below normal operating temperatures via the IG combination. Their analysis also suggested nondisordered arrangements of IL near the silica surfaces. In fact, Neouze and

Received: August 22, 2021

Revised: November 1, 2021

Published: November 17, 2021



Litschauer⁸ and others⁹ have observed structured arrangements of ILs near solid surfaces which clearly disrupt the bulk state of the ILs in confinement. Mitra et al.¹⁰ were able to examine the nature of phase transitions in IGs using neutron scattering. Guyomard-Lack et al.³ developed IG-like nanocomposite materials which they named “ionobrids.” For their Li-TFSI-Pyr1.3-based materials, they measured ionic conductivity over a range of temperatures. The resulting Arrhenius plots of conductivity showed that their IG-like materials could achieve conductivities comparable to the base electrolyte while extending the liquid range. A number of combined experimental and simulation investigations^{11–13} have measured bulk transport properties and have used simulation to illuminate transport mechanisms such as “vehicular” charge transport (where the Li⁺ solvation shell remains intact as opposed to change coordinating ions over time). Others^{14,15} have investigated the connection between ion solvation structure and transport. Using molecular simulation of slit pores, Ori et al.⁴ found that the local structure of a confined model IL was dominated by electrostatic interactions and the surface to volume ratio of IL in the hydroxylated silica slit pore. They also observed multilayered IL structures with a characteristic thickness of ≈ 20 Å near the planar surfaces. Last, Foroutan et al.¹⁶ presented a review of investigations of water and ILs nanoconfined in zeolites, micelles, and carbon nanotubes and other carbon-based nanostructures.

In this work, we study a Li-Pyr-TFSI IL confined in nanoporous silica as a model system. We construct a variety of nanoporous morphologies spanning a wide range of porosity and pore sizes and augment these systems with slit pores that have effectively zero tortuosity. We compute the correlated transport properties: Li⁺ tracer and collective diffusion, and ionic conductivity in each of these systems and reference the estimates to those for the pure IL. The variations in the silica structure clearly affect the effective transport properties. We observe that confinement improves transport properties for these liquids at low temperature, among other effects.

In the following, we outline the methods we employed to construct and characterize the model systems and measure their transport properties. Both tasks rely on previous developments, particularly refs 13 and 17. Then, we present the estimated transport properties and compare conductivity, collective and tracer diffusivity for six IG permutations (high and low silica fraction nanoporous IGs with narrow or wide pores, as well as two planar slit pores) relative to the bulk IL. The confinement effects on transport are connected to changes in partial radial distribution functions (RDFs) and other structural aspects are discussed. Last, we conclude with a discussion of the results in the context of design of IGs and future work.

2. METHODS

We used molecular dynamics (MD) to investigate the transport properties of a representative IG and corresponding IL, specifically 0.5 M Li-TFSI salt dissolved in a pyrrolidinium (Pyr1.3)-bis(trifluoromethane)sulfonimide (TFSI) IL contained in a nanoporous silica matrix. This model system is composed of molecular Pyr1.3⁺ and TFSI[−] ions and atomic Li⁺ ions, together with a solid silica (SiO_x) matrix with hydroxylated surfaces. These are common constituents of ILs. Li⁺ ions are widely used in electrolytes for batteries and other electrochemical devices. Pyr1.*x*⁺ molecules are composed of a carbon and nitrogen ring where the ring nitrogen is also

connected to a methyl carbon and an alkyl ligand of length *x*, refer to Figure 1a. The nitrogen in the ring is positively

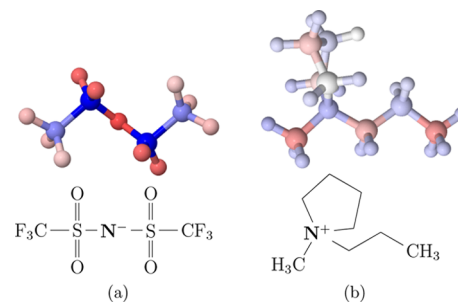


Figure 1. Molecular ions: (a) TFSI[−] and (b) Pyr1.3⁺. Atoms colored by partial charge (red: −1, blue: +1).

charged. TFSI[−] is a sulfur- and fluorine-containing molecule, refer to Figure 1b, with a center negatively charged nitrogen which is commonly used as a counterion in electrolytes because of its stability, ready dissociation of the Li-TFSI salt, and ionic conductivity. Silica was chosen as the confining matrix due its ubiquity and the availability of suitable and reliable potential parameters.

2.1. System Construction. The molecular model for classical MD of the IL and IGs is based on an empirical potential composed of short-range steric repulsion and van der Waals/dispersion, long-range Coulombic, and intramolecular covalent interactions. For the van der Waals and the Coulomb forces, the interaction energy is dependent on the atomic positions, through pairwise distances, and point charges. As is commonly practiced, the covalent intramolecular interactions maintaining the conformation of the individual molecules were modeled with harmonic potentials for pair bond distances and for three atom angles and a trigonometric potential for four atom dihedral angles (torsions). All parameters were taken from the literature. The parameters for silica and its hydroxylation are from Du and Cormack ref 18. As described fully in ref 13, the AMBER unified potential parameterization^{19–21} was used for IL components. The silica-IL interactions were taken from Cruz-Chu et al.²² Point charges for the molecular ions are from ab initio calculations in the previous study of a variety of Pyr ILs.¹³ As in the previous studies,^{13,17} a particle–particle particle–mesh²³ method was employed to compute the long-range Coulomb interactions beyond a 16 Å cutoff and direct summations were used for the remainder. Further details of the parameterization are in refs 13 and 17.

The process of constructing and equilibrating the IG systems is detailed in ref 17. We briefly outline it here. The Li-Pyr-TFSI IL was prepared according to the procedure described in ref 13 which involves a sequence of relaxation and isothermal-isobaric annealing steps starting from an expanded, low-density configuration. The silica generation process was similar in that its melting–annealing process was starting from a structured configuration (cristobalite). A negative pressure rupturing procedure²⁴ was used to generate nanoporous structures, where low/high speed rupturing created narrow/wide pores. Slow expansion rates and longer annealing between expansion steps allowed for more long-range relaxation of the silica. Two schedules were employed to generate narrow (cycles of 0.1 nm/5 fs expansion and 100 fs annealing) and wide pores (cycles of 1.0 nm/100 fs expansion and 1000 fs

Table 1. Systems: Pure IL (0.5 M Li Pyr1.3-TFSI) and Silica IG in Slit and Nanoporous Configurations^a

system	N	N_f	N_{Li}	N_{Pyr}	V [nm ³]	V_f [nm ³]	ζ [nm ⁻¹]	η [nm]	τ
IL	90,816	90,816	300	2048	929	929	0.000		1.00
60 Å slit	79,103	52,420	163	1186	865	502	0.363	3.000	1.00
25 Å slit	72,095	22,568	77	508	879	258	0.723	1.291	1.01
8% wide	79,263	68,951	271	1601	876	811	0.663	1.616	1.08
8% narrow	78,136	67,283	272	1498	866	816	0.516	1.010	1.08
35% wide	73,385	42,281	203	929	885	596	2.072	0.885	1.18
35% narrow	61,753	28,688	197	608	839	579	1.156	0.498	1.23

^aThe 8% and 35% silica fractions of the nanoporous systems are by volume. The total number of atoms in the system is denoted by N , whereas the number of atoms in the IL is N_f . Likewise, N_{Li} is the number of Li⁺ ions and N_{Pyr} is the number of Pyr ions. Number of TFSI⁻ counterions is determined by charge balance in the IL: $N_{TFSI} = N_{Li} + N_{Pyr}$. The system volume is denoted by V . The volume of fluid V_f and surface area are determined by probe sphere analysis using a sphere of radius 0.5 nm. The specific surface area ζ is the exposed silica surface area divided by V_f . Narrowness η is defined as the probe sphere radius at half the system's volume fraction. The tortuosity τ was evaluated with 0.25 nm tracer particle.

annealing). In addition to nanoporous configurations, we constructed slit pore IGs with the same silica and IL components for comparison. Confinement effects in these simple/more regular geometries were investigated by using two different slit widths: 25 and 60 Å. For each system, undercoordinated Si and O atoms on the silica surface were capped to form silanol surface terminations. To form the IG, the IL and silica components were intersected, the IL molecules overlapping the silica were deleted, and the resulting system was relaxed and annealed with isothermal-isobaric dynamics to a selected temperature T and zero pressure. All resulting systems have comparable volumes and number of atoms, refer to Table 1. Note that other work²⁵ has not reported significant size effects in transport estimates (described in the next section), so we expect that the systems are on a comparable basis.

The synthesis nanoporous silica aerogel can be tuned to effect different morphologies, leading to potentially different impacts on transport parameters. Figure 2 shows the model nanoporous systems with different silica volume fractions and pore morphologies. The main effect on transport, the IL volume fraction, is evident in comparing the 8% silica systems (Figure 2a,b) to the 35% silica systems (Figure 2c,d). The slower expansion schedule creates fewer and larger silica clusters and, correspondingly, larger pore diameters, which is clear in the contrast in the high-silica IGs (Figure 2c vs 2d), and slightly more subtle in the low-silica IGs (Figure 2a vs 2b). More evidence of this distinction will be given in the discussion of Figure 3 in the next section. The slit pores shown in Figure 2d,e were constructed for comparison of the nanoporous IGs to low tortuosity IGs and have relatively smooth surfaces. Also evident in all six systems is the clustering of TFSI⁻ and Pyr1.3⁺. The smaller pore sizes and slit width IGs are evidently small enough to disrupt these solvation structures, an effect which will be investigated in Section 4.

2.2. System Characterization. To influence the IG design, we would like to associate changes in transport efficiency with quantifiable structural characteristics. The metrics for structural characterization of a nanoporous solid are myriad.²⁶ Here, we focus on the controllable aspects that will likely affect the ionic transport, such as silica fraction and pore size. Table 1 gives the general statistics of all the systems we examined. Because we concentrate on ionic transport, the listed properties focus on the IL and the silica surface interactions with IL. The volume of fluid (V_f) and exposed surface area per fluid volume (ζ) are derived from probe sphere analysis.^{17,27,28} The silica fraction reported in Table 1 is

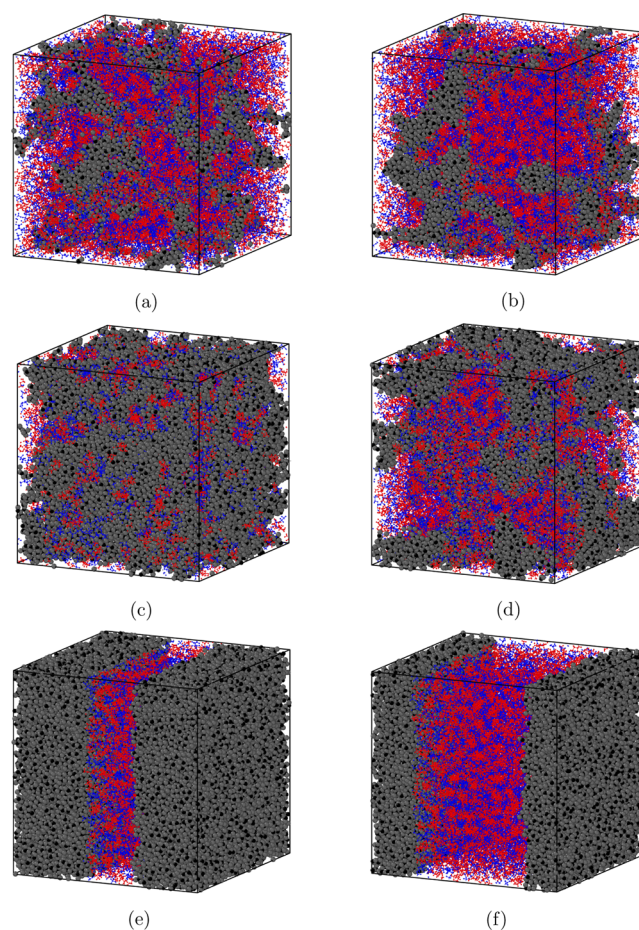


Figure 2. IG structures: (a) wide pore 8% silica, (b) narrow pore 8% silica, (c) narrow pore 35% silica, (d) wide pore 35% silica, (e) 25 Å wide slit pore, and (f) 60 Å wide slit pore. Black: Si, dark gray: O, light gray: H, red: Pyr1.3, blue: TFSI, and purple: Li.

also on a per volume basis and is defined as the complement of the fluid volume fraction, that is, $1 - V_f/V$. Note that the volume fraction and surface area both are probe sphere-dependent (the values in Table 1 are for a 5 Å probe sphere). The system sizes vary due to the variation in the structure and composition together with the equilibration using zero-pressure isobaric, isothermal dynamics. The Li molarities also vary due to small system sizes relative to molecule sizes. The tortuosity, defined as ratio of the trajectory path-length to the straight-line transit, was calculated using tracer particles

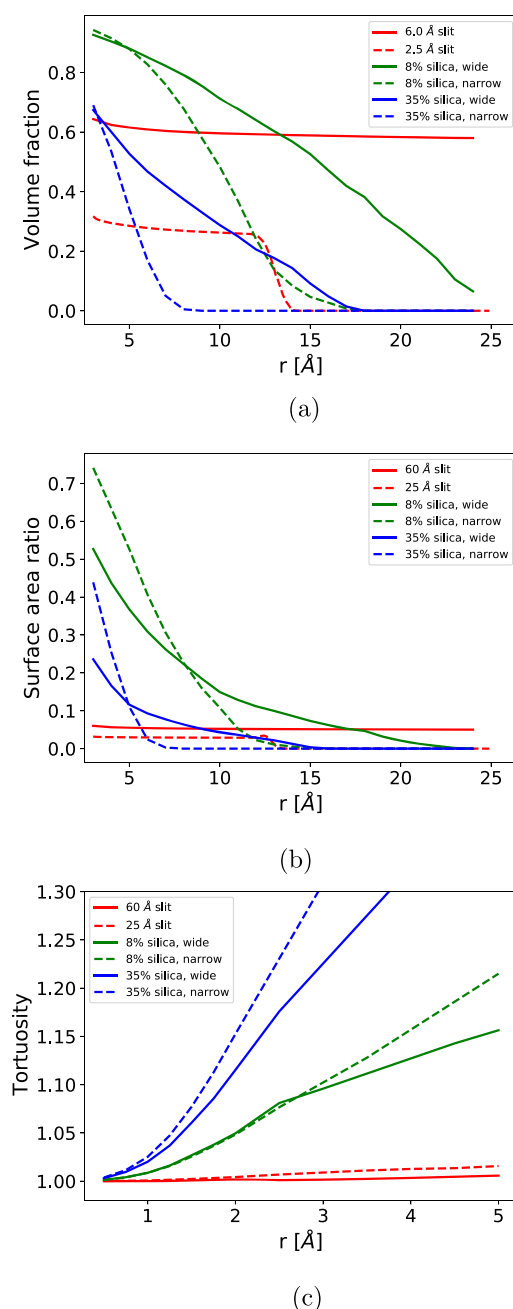


Figure 3. Structural characterization: (a) accessible pore volume distributions as functions of probe sphere radius, with (b) the surface area of the pore space calculated at each probe sphere radius, normalized by the accessible volume, and (c) tortuosity as a function of tracer particle radius.

interacting with the silica matrix using a Lennard-Jones (LJ) potential that provided a well-defined soft sphere radius.

We defined tortuosity τ as the ratio of the trajectory path length of tracer particles to the transit in the direction $\mathbf{e}$ of a small bias force

$$\tau = \left\langle \frac{\sum_{i=1}^n \|\Delta \mathbf{x}_i\|}{(\mathbf{x}_n - \mathbf{x}_0) \cdot \mathbf{e}} \right\rangle$$

where $\mathbf{x}_i$ is the position of a particular tracer at time-step i , and $\Delta \mathbf{x}_i = \mathbf{x}_i - \mathbf{x}_{i-1}$ is a displacement. Each tracer follows biased, damped dynamics

$$m\ddot{\mathbf{x}} = \mathbf{f} + \epsilon \mathbf{e} - \gamma \mathbf{v}$$

where $m = 1.0$ amu is an assigned mass, $\mathbf{f}$ is the interatomic force, $\epsilon = 100.0$ eV/Å is the magnitude of the bias force, $\gamma = 1.5$ eV ps/Å² is the velocity damping parameter, and $\mathbf{v} = \dot{\mathbf{x}}$ is the particle velocity. A tracer particle only interacts with the silica through $\mathbf{f}$, not with other tracers. Notably, in the absence of significant impacts, the steady-state velocity is $\mathbf{v} = \epsilon/\gamma \mathbf{e}$, which assists in choosing appropriate ϵ and γ . For convenience, we employed the initial locations of the Li⁺ ions for the tracer particles because they were already in the pore spaces and we froze the positions of the silica matrix to speed up the calculation. For the silica:tracer force $\mathbf{f}$, we chose a LJ interaction potential

$$\varphi(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

because it provides a well-defined soft radius $R = 2^{1/6}\sigma$. For the silica, the LJ energy ϵ was set to 0.01 eV and the LJ distance parameter σ was set to 3.5 Å, both of which are on the order of accepted values for argon. For the tracer particles, the same ϵ was used and σ was varied to affect a range of particle sizes in independent simulations. Lorentz–Berthelot mixing rules [$\sigma_{ab} = 1/2(\sigma_a + \sigma_b)$ and $\epsilon_{ab} = \sqrt{\epsilon_a + \epsilon_b}$] were employed to obtain the silica–tracer particle interaction parameters (note that no electrostatic or other interatomic interactions influenced the tracer particle dynamics.)

Figure 3a shows the distribution of volume accessible by a probe sphere of a given size, and Figure 3b shows the corresponding distribution of the accessible surface area per volume for the same probe spheres. The pore size distributions of the model nanoporous systems are comparable to pores in actual silica IGs, which are reported to be 1–30 nm on average.^{29–31} Note that the limit of V_f for small radius probe spheres (but larger than interatomic spacing) is determined by the chosen silica fraction. The wider pore systems are characterized by significantly higher and broader volume and specific surface area distributions. The narrowest nanoporous IG (35% silica with narrow pores, Figure 2c) has no accessible volume above $r \approx 8$ Å while the widest (8% silica with wide pores, Figure 2b) has accessible volume to ≈ 20 Å. The width of the narrow slit system falls in the center (half width 12.5 Å) of the range of the nanoporous pore size distributions, while the wider slit has an aperture well above the typical pore size (half width 30 Å). Note that it is apparent from Figure 3a that even the slits have fine-scale features. The initial slow decay of the accessible volume curve is primarily due to the surface hydroxyl groups, whereas the subsequent rapid decay is due to the larger scale roughness of the nominally flat slit walls. As can be seen in Figure 3b, the surface area per accessible volume fraction is higher for the low-silica nanoporous IGs (relative to high-silica IGs), indicating a rougher and more convoluted surface. Likewise, the specific surface area is greater for the wide pore systems, especially at large probe sphere radius. However, it appears at the smallest probe sphere radii, which can be taken as characterizing the entire pore space, and the surface area per accessible volume is marginally greater for the narrow pore systems than for their wide pore counterparts. The tortuosity as a function of tracer particle radius shown in Figure 3c displays the expected trends and ordering, for example, the tortuosity increases with tracer particle radius and the slits have effectively no tortuosity $\tau \approx 1$. The narrower slit

has a slight rise in τ with the increase in particle size as the larger particles interact with the walls more frequently. Likewise, the narrower IGs have higher tortuosity than their wider counterparts, with the difference between the high-silica IGs being more distinct than the low-silica IGs.

2.3. Transport Coefficients. We used well-known relations to extract the transport coefficients from equilibrated systems (refer to refs 32–36). Ionic conductivity σ can be estimated with a Green–Kubo (GK) relation

$$\begin{aligned}\sigma &= \frac{1}{3 V k_B T} \int_0^\infty \langle \mathbf{I}(0) \cdot \mathbf{I}(t) \rangle dt \\ &= \frac{1}{3 V k_B T} \int_0^\infty \sum_\alpha \sum_\beta q_\alpha \mathbf{v}_\alpha(0) \cdot q_\beta \mathbf{v}_\beta(t) dt\end{aligned}\quad (1)$$

which relies on the time correlation of the ionic current $\mathbf{I}(t) = \sum_\alpha q_\alpha \mathbf{v}_\alpha(t)$ where q_α is the charge and $\mathbf{v}_\alpha$ is the velocity of atom α . Here, k_B is Boltzmann's constant and V is the system volume. Likewise, the per-species currents $\mathbf{J}_a(t) = \sum_{\alpha \in \mathcal{G}_a} \mathbf{v}_\alpha(t)$ lead to the (Onsager, collective) diffusivity coefficients $D_{a:b}$ ³⁵

$$\begin{aligned}D_{a:b} &= \frac{1}{3 N} \int_0^\infty \langle \mathbf{J}_a(0) \cdot \mathbf{J}_b(t) \rangle dt \\ &= \frac{1}{3 N} \int_0^\infty \sum_{\alpha \in \mathcal{G}_a} \sum_{\beta \in \mathcal{G}_b} \mathbf{v}_\alpha(0) \cdot \mathbf{v}_\beta(t) dt\end{aligned}\quad (2)$$

where N is the total number of atoms and $\mathcal{G}_a$ is the group of atoms belonging to chemical species a . Note: (a) the Onsager coefficients relate the diffusion flux to the gradient in the chemical potential (not the Fickian coefficients associated with the concentration gradient^{34–36}), and (b) this form of the GK relation is similar to the formula for tracer diffusion eq 3 but omits the $1/k_B T$ factor in eq 1. Alternate forms of this GK relation exist.³⁴ The diffusion coefficient d_a for the tracer diffusion of species a is given by the Einstein relation

$$d_a = \frac{1}{3} \int_0^\infty \langle \mathbf{v}(0) \cdot \mathbf{v}(t) \rangle dt = \frac{1}{3 N_a} \int_0^\infty \sum_{\alpha \in \mathcal{G}_a} \mathbf{v}_\alpha(0) \cdot \mathbf{v}_\alpha(t) dt\quad (3)$$

where N_a is the number of atoms in $\mathcal{G}_a$.

These measures of ionic transport are correlated and related in the nested sense. The conductivity σ has solid–fluid velocity correlations, and the collective diffusion of the fluid components $D_{a:b}$ has velocity correlations of the same species, whereas the tracer diffusion d_a only has flux correlations of the same molecule or atom. The dependence of these relations is particularly clear for the monoatomic species, Li^+ . The collective diffusion of Li^+ , $D_{\text{Li:Li}}$, can be separated from the other contributions to the ionic conductivity

$$\begin{aligned}\sigma &= \frac{1}{3 V k_B T} \left(e^2 D_{\text{Li:Li}} \right. \\ &\quad \left. + \sum_{a \neq \text{Li}} \sum_b \int_0^\infty \sum_{\alpha \in \mathcal{G}_a} \sum_{\beta \in \mathcal{G}_b} q_\alpha q_\beta \mathbf{v}_\alpha(0) \cdot \mathbf{v}_\beta(t) dt \right)\end{aligned}\quad (4)$$

because the charge of Li^+ is e . Note that the ratio $\sigma/D_{\text{Li:Li}}$ is related to the transference number $I/(eJ_{\text{Li}})$ typically used to assess how much of the ionic current is due to a particular ionic species, in this case Li^+ ; however, this ratio does not include the contributions from the correlations of Li^+ with the

other ionic species. Similarly, the collective diffusion coefficient $D_{\text{Li:Li}}$ includes the tracer diffusion coefficient d_{Li}

$$D_{\text{Li:Li}} = \frac{N_{\text{Li}}}{N} d_{\text{Li}} + \frac{1}{3 N} \int_0^\infty \sum_{\alpha \in \mathcal{G}_{\text{Li}}} \sum_{\beta \neq \alpha \in \mathcal{G}_{\text{Li}}} \mathbf{v}_\alpha(0) \cdot \mathbf{v}_\beta(t) dt\quad (5)$$

with these two diffusion measures, $D_{\text{Li:Li}}$ and d_{Li} , the main difference is that $D_{\text{Li:Li}}$ includes the cross-correlation of velocities of different atoms.

These transport measures provide a compact quantification of correlated effects with well-established properties. To present the differences in transport in IGs, baselined to the pure IL, beyond the main effect of fluid volume fraction, we employ the same coefficients on a volume of fluid basis

$$\tilde{\sigma} = \frac{1}{3 V_f k_B T} \int_0^\infty \sum_\alpha \sum_\beta q_\alpha \mathbf{v}_\alpha(0) \cdot q_\beta \mathbf{v}_\beta(t) dt\quad (6)$$

and

$$\tilde{D}_{\text{Li:Li}} = \frac{1}{3 N_{\text{Li}}} \int_0^\infty \sum_{\alpha \in \mathcal{G}_{\text{Li}}} \sum_{\beta \in \mathcal{G}_{\text{Li}}} \mathbf{v}_\alpha(0) \cdot \mathbf{v}_\beta(t) dt\quad (7)$$

where the cell volume V has been replaced by the volume of fluid V_f and the total number of atoms N has been replaced by the number of Li^+ ions N_{Li} . This renormalization to obtain properties intrinsic to the fluid also removes the ambiguity of applying the GK formulas to the slit system, that is, silica slab thickness is arbitrary and does not affect the fluxes but does affect the system volume and total number of atoms. Also, because the flux in the slit systems is clearly restricted to the two lateral directions we averaged the correlations over only these two directions (instead of three in the other systems).

After pressure and temperature equilibration (described in the previous section), flux data were collected with isothermal dynamics (time step 0.5 fs, duration 5.0 ns) every 5.0 fs for a total correlation interval of 50 ps (the upper limit of the integrals³⁷) and was averaged over 10^6 samples. Convergence of the transport properties was monitored by the integrals, eqs 3, 6, and 7, evaluated with trapezoidal quadrature.

3. RESULTS

Using the model IG systems with different morphologies, we explore the sensitivity of ionic conductivity, σ , and the collective and tracer diffusion coefficients of Li^+ , $D_{\text{Li:Li}}$ and d_{Li} , respectively, to variations in temperature, overall porosity, and pore size. We use the pure IL as a baseline for comparison and report the transport properties on a volume of fluid basis (as described in the previous section). This comparison on an intrinsic (volume of fluid) basis removes the first-order IL fraction effect (i.e., more IL per volume leads to higher transport coefficients) and reveals more subtle sensitivities. We link these results to structural characteristics of the different IG configurations.

For reference, the ionic conductivity for this IL, $\sigma = 0.0887$ S/m, was measured experimentally in previous work.¹³ The Li^+ diffusivity of a lower molarity, related IL (0.1 M Li-Pyr1.4) was also measured experimentally, $0.009 \mu\text{m}^2/\text{ms}$, in ref 38. Previous work¹³ also provided experimental measurements, the viscosity of the 0.5 M Li-Pyr1.3 IL studied in this work, 145.69 cP; however, viscosity is difficult to estimate with MD due to the longer characteristic relaxation times.²⁵ Gittleston et al.¹³ estimated the transport properties of the model IL

employed herein and compared them to experimental measurements which provides some validation of the potential used in this work.

3.1. Transport Coefficients. Figure 4 shows the Arrhenius behavior of the following: (a) the Li^+ tracer diffusivity, d_{Li} (b)

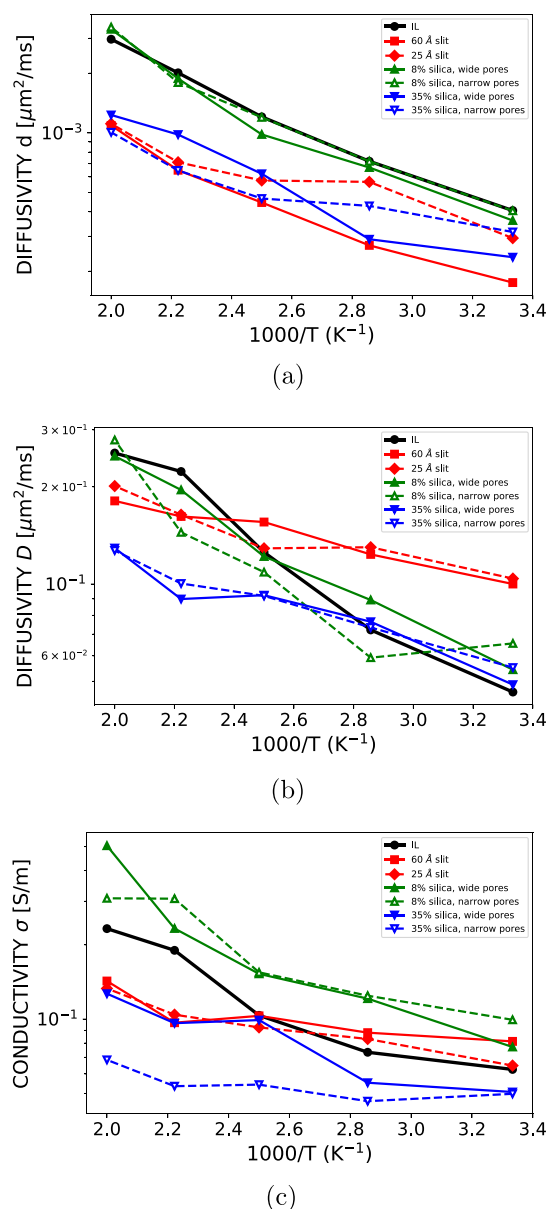


Figure 4. Arrhenius behavior of: (a) the tracer diffusivity d_{Li} (b) the collective diffusivity $D_{\text{Li:Li}}$ and (c) the ionic conductivity σ .

the Li collective diffusivity, $D_{\text{Li:Li}}$, and (c) the ionic conductivity, σ , for all six IG configurations listed in Table 1 and the baseline pure IL.

The tracer diffusion, d_{Li} shown in Figure 4a, is the property most specific to the transport of isolated Li^+ ions. As can be seen, the trends for all the IGs resemble the IL baseline and clearly shows the asymptotic approach of IG to IL with fluid volume fraction. In fact, the low-silica IG Arrhenius trend is nearly identical to the IL on a per volume of fluid basis. The slit pores have the lowest d_{Li} , while the high-silica IGs have slightly larger d_{Li} . The narrow low-silica IG follows the IL trend closely; however, the high-silica narrow pore IG and the

narrow slit display the pronounced deviation from a (slightly upward curved) $\log d_{\text{Li}}$ versus inverse temperature trend, particularly at low temperatures. This flattening of the diffusivity–temperature trend appears to be a confinement effect, as it is most pronounced at higher silica fractions.

The collective diffusion, $D_{\text{Li:Li}}$ shown in Figure 4b, incorporates cross Li^+ ion correlation effects which apparently tend to flatten the trends with temperature for the high-silica IGs and the slit pores. Again, low-silica IGs generally follow the IL trend closely. Interestingly, for this transport property, the slit pores have intermediate $D_{\text{Li:Li}}$ at high temperatures and distinctly higher $D_{\text{Li:Li}}$ at low temperatures. Comparing narrow and wide pore systems at the same silica fraction indicates that the pore width has a minimal effect on $D_{\text{Li:Li}}$.

Finally, the ionic conductivity, σ shown in Figure 4c, includes contributions from the other IL ions (Pyr^+ and TFSI^-). Here, all the IGs display flatter temperature trends than the pure IL. This reduced temperature sensitivity has clear technological benefits. On a per fluid volume basis, the low-silica IGs outperform not only the other IGs but also the IL. Little difference is seen between the narrow and wide variants at this silica fraction. Likewise, the narrow and wide slits have similar conductivities, which are between the values for the high and low-silica IGs. The high-silica IGs, in contrast, display distinct conductivities for the narrow and wide pores, with ionic conductivity for the wide pore variant being on par with the slits. This may be due to a structured surface layer of ions that is able to form on the atomically flat slit pores (the interplay of structure and transport will be discussed more fully in the next section).

In summary, we observe that the Li^+ tracer diffusion in IGs approaches the IL trend as the silica fraction decreases, whereas there is enhanced behavior seen in the collective transport properties. In fact, there are indications that the ionic conductivity of low-silica fraction IGs can be high enough on a per fluid volume basis to offset the first-order IL fraction discount. Furthermore, we observe trends with temperature in high-silica IGs that are suggestive of confinement effects. Generally there is a flattening of the temperature dependence for the collective transport properties at higher silica fractions. Similarly, the tracer diffusion shows deviations from the IL trend but only for narrow pore, high-silica IGs, and the narrow slit IG. The different behaviors displayed by the three transport properties is largely due to the fact that Li^+ is small and mobile relative to the other charge carriers $\text{Pyr}^{1.3+}$ and TFSI^- . Comparing the ionic conductivity to the Li^+ diffusivity, it is clear that these larger ions play a significant role in conductivity. Their larger size may account for the relatively higher sensitivity of the ionic conductivity to pore geometry. Note that the uncertainties in the transport estimates due to undersampling replica morphologies may be substantial. We could only afford one sample per controllable gross structural characteristics because each took approximately 50,000 cpu h. We spaced the selected porosity and pore widths of the systems widely in order for the differences in the calculated transport coefficients to be greater than the sampling variation estimated in preliminary studies.

These qualitative observations are quantified in Table 2 which summarizes fits of each of the trend lines to a simple Arrhenius relation $X = X_0 \exp(-E_X/k_B T)$, where $X \in \{d, D, \sigma\}$ and k_B is Boltzmann's constant. This table provides a means of ranking the IGs (despite the moderate deviations from a simple Arrhenius behavior) that will be explored further in the

Table 2. Arrhenius Fits of Transport Coefficients^a

system	$\log d_0$	E_d	$\log D_0$	E_D	$\log \sigma_0$	E_σ
IL	-2.89	12.4	1.39	11.2	0.54	8.7
60 Å slit	-4.29	11.1	-0.81	3.8	-1.40	2.9
25 Å slit	-5.18	7.2	-0.78	3.8	-1.10	3.9
8% wide	-2.65	13.5	0.87	9.6	1.56	10.6
8% narrow	-2.79	12.8	0.58	9.0	0.69	7.7
35% wide	-4.03	11.1	-0.83	5.3	-0.67	5.9
35% narrow	-5.50	6.6	-0.91	5.0	-2.37	1.7

^aThe units of the prefactors X_0 are as follows: d_{Li} : [$\mu\text{m}^2/\text{s}$], $D_{Li:Li}$: [$\mu\text{m}^2/\text{s}$], and $\tilde{\sigma}$: [S/m]. Pseudo-activation energies E_X are given in [kJ/mol].

next section. For reference, Castiglione et al.³⁸ provides an estimate of $E_d = 46 \text{ kJ/mol}$ for Li^+ in a 1:9 mixture of Li:TFSI and Pyr1.4:TFSI.

4. DISCUSSION

Using RDFs and the structural characterization developed in Section 2.2, we attempted to relate the transport coefficients discussed in the previous section to changes in the matrix and liquid structure of the suite of IGs. Clearly, the transport coefficients, $\tilde{d}_{Li}$, $\tilde{D}_{Li:Li}$, and $\tilde{\sigma}$, vary with the matrix structure. Also, pore size is an important structural feature, with IGs that have narrower pores having lower transport coefficients, as can be seen in Figure 4. However, pseudo-activation energies (reported in Table 2) are often smaller for these systems, leading to changes in the relative transport efficiency of the different systems as the temperature changes. Experimental observations indicate confinement as the cause of the depression of the glass transition temperature^{3,39,40} and correspondingly lower IL viscosity. This motivates an investigation of the fluid structure to see if confinement effects inferred from the transport properties can be related to changes in the local structure of the confined IL.

With this objective, we characterized the structure of the liquid by computing (partial) RDFs for different species in the IL (Figures 5 and 6), as well as for IL species and atoms in the silica matrix (Figure 7). RDFs computed between species in the IL track variations in the structure of the liquid under confinement, while those between IL species and the solid

matrix reveal variations in the surface adsorption layer. Previous studies of ILs in atomically flat slit pores^{4,41–43} show that confined ILs can be roughly separated into two regions: a bulk-like region in the center of a pore and a surface-like region where ions and molecules are adsorbed to the surface of the matrix. Three RDFs are particularly informative for these purposes: (a) $\text{Li}^+:\text{TFSI N}^-$, (b) $\text{Pyr N}^+:\text{TFSI N}^-$, and (c) $\text{Li}^+:\text{surface silica O}^-$ (refer to Figure 1 for the location of N^- in the TFSI ion and N^+ in the Pyr ion, both are distinguished by bold font in the schematics). Time-averaging of RDFs over the course of the simulations is essential, particularly for IGs, as the transport of species between surface- and bulk-like regions will change their coordination environment. As can be seen in Figures 5–7, reducing the pore apertures causes distinctive structures to appear in the confined IL, both in the bulk-like environment and at the surface. There is some evidence that the ordering (or disruption of ordering relative to a flat slit geometry) may be species and size-dependent. It is likely that the near-surface ordering of the molecular IL species reported in ref 4 is disrupted in nanoporous systems due to the boundary layers being comparable to the curvature of the silica pores and that the smaller Li ion is less affected; however, more studies are needed to make confident conclusions.

Examining Figure 5a, the largest peak of the $\text{Li}^+:\text{N}^-$ RDF in the unconfined IL is located at $\approx 3.6 \text{ Å}$ with a secondary peak located at $\approx 4.3 \text{ Å}$. The first peak corresponds to the bidentate Li-TFSI conformation, in which the Li^+ ion is bonded to two O atoms in a TFSI molecule, while the smaller peak (at 4.3 Å) is associated with the monodentate Li-TFSI conformation, in which Li is bonded to a single O atom in a TFSI molecule, as identified in previous MD studies.^{44–46} This RDF captures the baseline structure of the Li^+ solvation shell, variations in which have significant consequences for Li^+ transport.¹³ Confining the IL in a silica matrix produces changes in the shape of the $\text{Li}^+:\text{N}^-$ RDF, Figure 5a and in its variation with temperature, Figure 6a. The overall shape of the $\text{Li}^+:\text{N}^-$ RDF for the 8% silica IGs, whether with narrow or wide pores, resembles that of the pure IL. However, the height of the monodentate peak, relative to the bidentate peak, is larger for more confined IGs and slit pores. The highest monodentate peaks are seen in

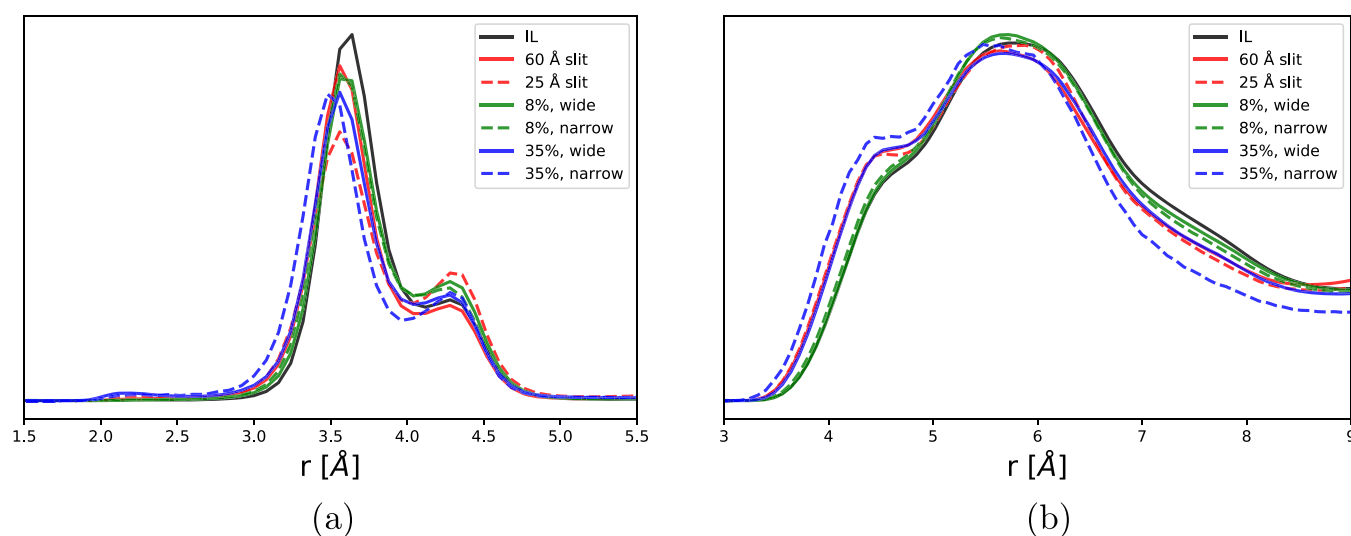


Figure 5. System dependence of the radial distribution of $\text{Li}^+:\text{TFSI N}^-$ and $\text{Pyr N}^+:\text{TFSI N}^-$ at 300 K.

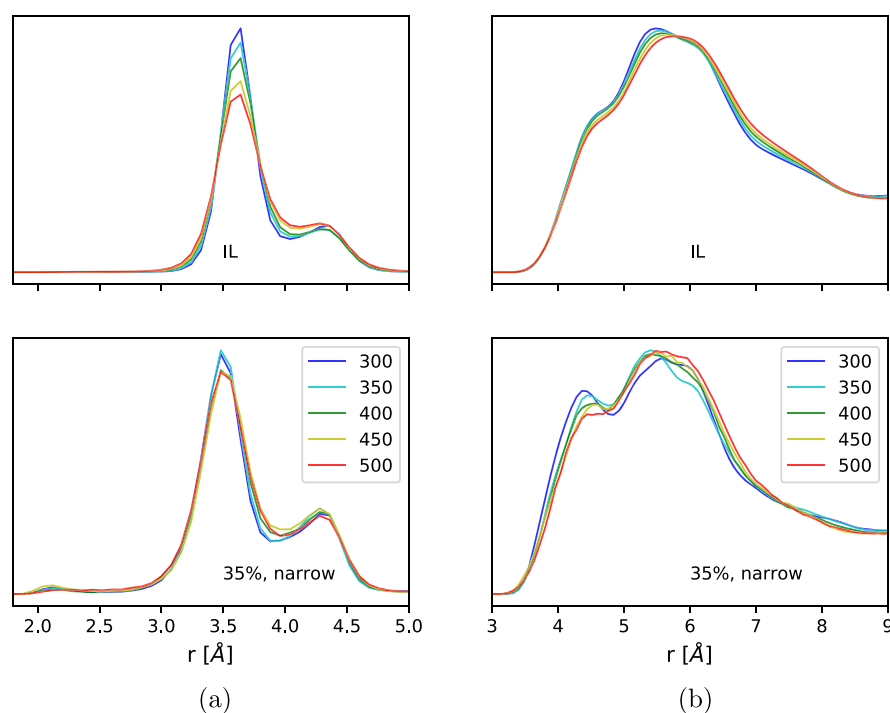


Figure 6. Temperature dependence of the radial distribution of: (a) $\text{Li}^+:\text{TFSI N}^-$ and (b) $\text{Pyr N}^+:\text{TFSI N}^-$.

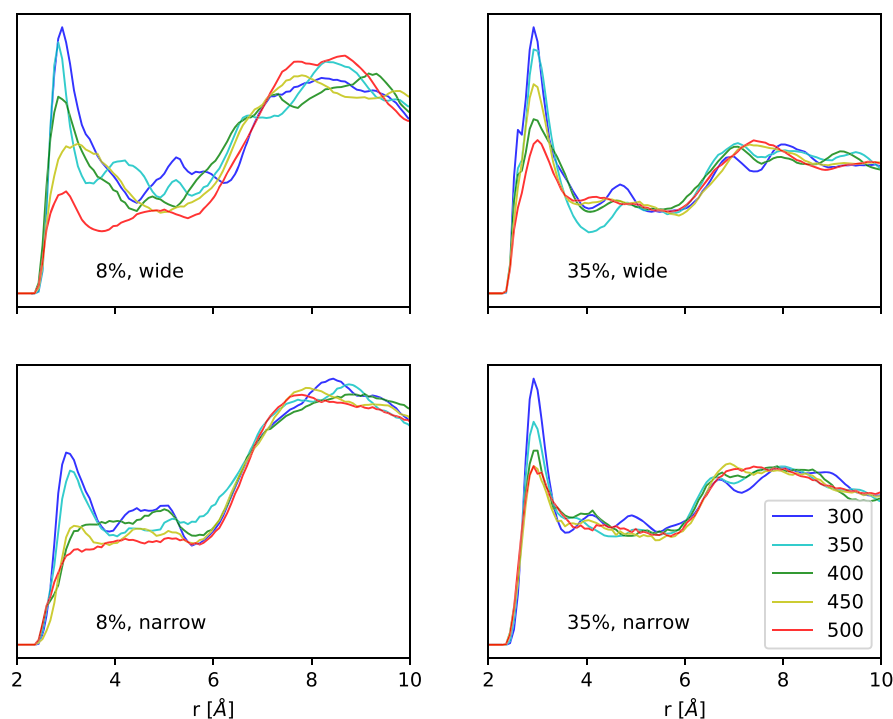


Figure 7. Temperature dependence of the radial distribution of IL Li^+ : surface silica O^- .

$\text{Li}^+:\text{N}^-$ RDFs for the 35% silica IGs and 25 Å slit, while the lowest are seen in the partial RDFs of the IL and 60 Å slit. Monodentate peaks for the 8% silica IGs are intermediate between these extremes. In an experimental investigation, Brinkkoetter et al.⁴⁷ found that relatively slow, “vehicular” Li^+ transport, where a large Li^+ solvation shell remains intact, is associated with the bidentate conformation, whereas the monodentate conformation corresponds to faster “structural” diffusion. This is connected with the fact that counter currents

of Li^+ complexes have been observed in related ILs^{47–50} which inhibit ionic transport. These observations appear to be related to the faster low-temperature Li diffusion in the more confined IGs in our study, as more confined systems have higher monodentate (and lower bidentate) peaks and are more likely to restrict the motion of large complexes (and promote alternative transport mechanisms).

The most confined systems, which are the 25 Å slit and both the narrow and wide pore 35% silica IGs, have an additional

minor peak at 2.2 Å, which disappears at elevated temperature. This corresponds to a rare $\text{Li}^+:\text{TFSI}^-$ conformation in which the orientation of the TFSI molecule is reversed relative to the bidentate conformation, bringing Li^+ and the N in the TFSI^- ion into close proximity. This additional $\text{Li}^+:\text{TFSI}^-$ conformation is correlated with confinement and likely corresponds to the increased packing efficiency of the Li^+ solvation clusters enforced by the matrix. Clearly, increasing temperature broadens these peaks (refer to Figure 6a). However, temperature also changes the relative heights of the two main peaks. While the height of the bidentate peak decreases with temperature, the height of the monodentate peak increases, indicating a shift in the prevalence of the two conformations. Absolute changes in the peak height with temperature are greatest for the IL and least for the narrow pore 35% silica IG. Temperature sensitivity of the Li transport is greatest for the IL and least for the narrow pore 35% silica IG, suggesting that changes in the Li^+ transport are related to temperature-induced changes in the Li^+ solvation shell. As mentioned, the additional/third peak seen in the most confined systems subsides at high temperatures. We conjecture that this is linked to the increased pore volume due to the expansion at high temperatures, which implies that the formation of the energetically unfavorable conformation is no longer driven by the geometric requirements for the efficient packing of the IL into a narrow pore space.

As is apparent from the difference in trends in Figure 4, the molecular components of the IL (Pyr1.3^+ and TFSI^-) contribute significantly (together with Li^+) to the total ionic conductivity. The $\text{Pyr N}^+:\text{TFSI N}^-$ RDF, shown in Figures 5b and 6b, can be used to assess the arrangement of the Pyr1.3^+ and TFSI^- molecular ions. In the pure IL, this RDF reaches a maximum at ≈ 5.3 Å at 300 K. There is a secondary peak at ≈ 4.5 Å that is partially obscured by the primary peak. The peaks broaden with temperature and shift to higher interatomic distances as the IL thermally expands. Examining Figure 5b, the $\text{N}^+:\text{N}^-$ RDFs for IGs and ILs in slit pores broadly resemble those of the unconfined IL. However, the secondary peak (at ≈ 4.5 Å at 300 K) has a higher relative height for confined ILs. This effect is least pronounced for the 8% silica IG and most pronounced for the narrow pore 35% silica IG (Figure 5b). Indeed, for this last system, as well as for the wide pore 35% silica IG and the 25 Å slit, the secondary peak is sufficiently high for a local maximum to be resolvable.

As illustrated in Figure 6b, the primary peak of the $\text{N}^+:\text{N}^-$ shifts to higher radii as the temperature increases. However, this shift is smaller for the IGs than the unconfined IL. This is a consequence of the stiff confining matrix, which must be dilated by thermal pressurization of the IL (and thermal self-expansion) for the fluid density to be able to decrease. As can be seen in Figure 6b, increasing temperature is associated with the decrease in relative height of the secondary peak at ≈ 4.5 Å (as was observed in Figure 6a). Indeed, for the narrow pore 35% silica IG, which is the most confined system, there is no longer a local maximum associated with this peak at 500 K. Further evidence that the appearance of this peak is driven by geometric requirements imposed by confinement within a narrow pore space is furnished by its disappearance at high temperature, as dilation of the pores during thermal expansion of the IG reduces the confinement experienced by the IL.

The appearance of additional peaks in the $\text{N}^+:\text{N}^-$ and $\text{Li}^+:\text{N}^-$ RDFs of the ILs with the narrowest pore spaces (as defined by the probe sphere analysis) indicates that these

conformations are produced by the confinement of the IL. These peaks are likely associated with higher packing efficiency of the IL molecules and $\text{Li}^+:\text{TFSI}^-$ solvation clusters. The appearance of additional peaks under confinement also provides evidence for local disruption of the bulk solvation of the IL. As observed by others,^{8,9} this type of disordering depresses IL glass transition temperatures, and this may explain the relatively higher transport coefficients for the slit pores and 35% silica IGs, especially the narrow pore 35% silica IG, at low temperature.

The IL:IL RDFs just discussed display the effect the confinement of the IL in the pore space of matrix has on the structure of the IL. In addition, direct interactions between the IL and the silica surface play a role in the transport. Figure 7 shows the computed surface silica $\text{O}^-:\text{IL Li}^+$ RDFs for the selected IGs across the entire temperature range (300–500 K) studied (note that the comparatively small surface areas of the slit pore systems, reported in Table 1, prevent the calculation of converged partial RDFs for this ion pair; however, they resemble those of the wide pore, low-silica IG). At 300 K, all the nanoporous IG $\text{O}^-:\text{Li}^+$ RDFs have a significant peak at ≈ 2.9 – 3.1 Å, which is associated with Li^+ ions adsorbed to the silica surface. At intermediate distances (3.5–5.5 Å), the $\text{O}^-:\text{Li}^+$ RDFs decrease significantly after this peak. In this range, the prevalence of Pyr1.3^+ appears to dominate (as evidenced by the $\text{O}^-:\text{N}^+$ RDFs, not shown for brevity), indicating a (semi-structured) surface packing based on the size of the cations. The TFSI^- counterion also becomes prevalent in this range. Further from the surface, the $\text{O}^-:\text{Li}^+$ RDFs for all systems increase to the bulk value. For the 8% silica IGs, the height of the adsorption peak is comparable to or lower than the bulk value; however, the adsorption peak is higher than the bulk value for the 35% silica IGs, perhaps reflecting the greater total volume of the IL that is close to the silica surface and a more surface-like rather than bulk-like environment.

The surface-like region of the IL may be reasonably defined as the volume extending from the surface out to the start of the bulk-like section of the $\text{O}^-:\text{Li}^+$ RDF, which is ≈ 6 Å from the surface. The difference between the 8 and 35% silica IGs can then be understood with reference to Figure 3a. A 6 Å radius probe sphere finds much of the pore volume of the 8% silica IGs to be accessible, whereas only a small fraction of the pore volume for the 35% silica IGs is accessible to a sphere of this size. The smaller fraction of the fluid that is bulk-like may explain the slower transport properties of the high-silica IGs, as transport properties are likely lower in the surface-like region of the IL. Indeed, across all systems, there is a modest positive correlation (0.3–0.5) between the specific surface area ζ and the pseudo-activation energy for the three transport properties we examined.

As seen in Figure 7, the temperature sensitivity for the $\text{O}^-:\text{Li}^+$ adsorption peak is more pronounced than those of the IL components. In particular, the height of the first $\text{O}^-:\text{Li}^+$ RDF peak decreases with the increase in temperature. This is consistent with thermally activated desorption of the surface-adsorbed Li^+ and their dissolution into the bulk-like region of the IL. The adsorption peak for the 8% silica IGs almost completely disappears at 500 K. Although it has not disappeared in the 35% silica IGs, the relative peak height has decreased to bulk-like values. Desorption may explain the nonlinear relationship between the logarithm of Li^+ diffusivity and the reciprocal temperature because in addition to Li^+

diffusion being intrinsically faster at high temperatures, there is also an increasing contribution from the relatively faster bulk transport due to Li^+ desorption from the silica surface.

Last, we have some evidence of segregation of the ionic species of the IL^{51,52} in smaller pores according to ion size. Figure 8 shows the distribution of IL coordination environ-

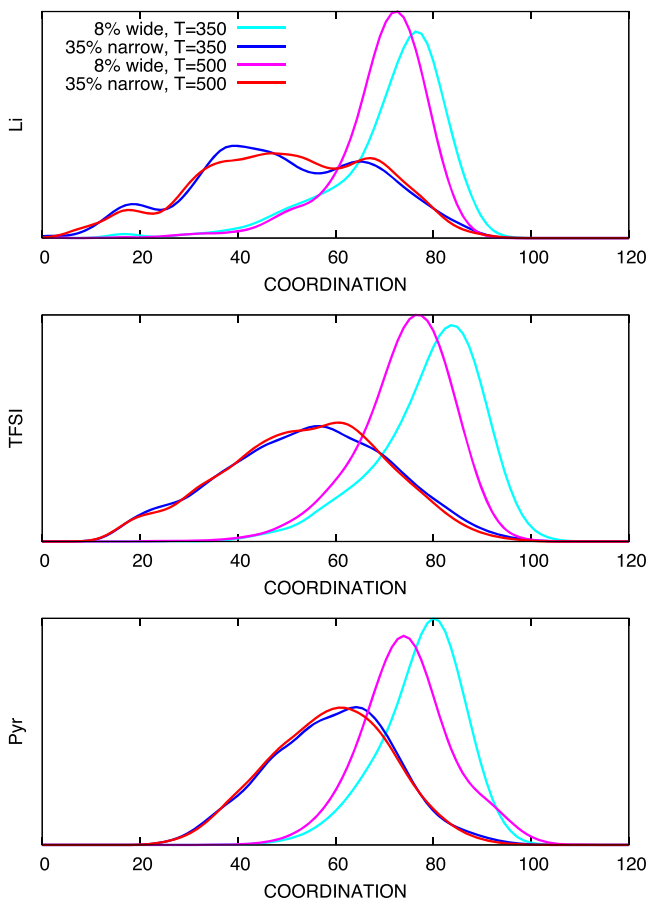


Figure 8. IL coordination distributions for the three components of the IL for the least- and most-confined IGs.

ments for the three components of the IL for the most- (35% silica, narrow pores) and least-confined (8% silica, wide pores) IGs. The coordination is a proxy for pore volume surrounding the ions because the silica matrix atoms were excluded from the 6 Å radius spherical sample volumes around the ions. From this data, a number of effects can be observed and the degree of apparent ionic segregation provides another structural distinction between IG morphologies. It is obvious that the major peaks in the coordination number are ordered in the reverse molecule size, that is, smallest ion peaks first. More importantly, there is a second trend based on ion size, the smaller ions, Li^+ and to a lesser extent TFSI^- , display a secondary peak at the low end of coordination in the most confined environments which is essentially absent for Pyr^+ . The pronounced existence of this peak in the Li^+ coordination probability density for the narrow, 35% silica IG points to the presence of Li^+ in the narrowest pores. The relatively small perturbation of this peak for the wide, 8% silica IG implies that this is likely not associated with surface segregation without confinement. The perturbation is clearly observable for TFSI^- in the narrow pore IG but is completely absent in the wide pore 8% silica system. No subsidiary peaks can be discerned for

the Pyr^+ charge center. From this evidence, we conclude that the Pyr^+ is essentially in a bulk environment and likely segregated in the larger pores, Li^+ is present in the smallest pores, and TFSI^- is in an intermediate state. It appears that these distributions are relatively stable with temperature (the shift observed for wide, 8% silica IG is attributable to the significant thermal expansion of that IG).

5. CONCLUSIONS

Using combinations of low and high silica volume fraction and pore size together with slit pore and pure IL systems, we investigated the effects of pore morphology on Li^+ tracer and collective diffusion and overall ionic conductivity. All the model IGs had significantly flatter temperature dependence of the transport coefficients than the corresponding pure IL, indicating a reduced tendency to freeze out. In general, low-silica, wide pore IGs displayed the best transport properties, but the trends with morphology for the nested set of transport coefficients changed as a collective behavior of the Li^+ ions and the molecular IL components were considered. Remarkably, the less-confined model IGs can have better collective transport properties on a volume of fluid basis than the constituent IL. We were able to qualitatively tie these trends to structural changes revealed by the RDFs of the IL components and the silica surface. For instance, the decreasing adsorption peak in the surface silica $\text{O}^-:\text{IL Li}^+$ RDF appears to predict the nonlinearity of the transport Arrhenius behavior for this system because dissolution/desorption of Li^+ in the confined IL changes the relative contributions from bulk-like and surface-like. The greater nonlinearity of the Arrhenius plots for the transport properties of high-silica IGs can be related to their higher surface area to pore fluid ratios, so that the adsorption of Li^+ ions to the silica surface has a greater impact on the number of ions dissolved in the bulk-like IL.

This investigation suggests a number of avenues for future work. The sampling of the structural characteristics we could explore was limited due to the computational cost of obtaining long-time flux correlations for the low/high volume fraction and narrow/wide pore IGs we examined. A larger sample of IG structures should reveal higher correlations with measurable and controllable structural characteristics of the IGs. Furthermore, in the literature, the use of GK methods for heterogeneous and, in particular, fluid–solid systems has been limited. To our knowledge, most work has focused on fluid slabs/slit pores (with some work examining walls with texture). Bocquet and Barrat⁵³ developed a GK formula for wall friction analogous to bulk GK formulas and applied to a fluid slab interacting with simplified walls. Priezjev and Troian⁵⁴ applied the Bocquet and Barrat formula and compared to other methods such as direct simulation of Couette flow. Later, Huang and Szulfarska⁵⁵ proposed an alternative formulation that apparently addressed shortcomings in the Bocquet and Barrat method. Nakaoka et al.⁵⁶ applied both the Bocquet and Barrat, and Huang and Szulfarska formulations to a water–methanol system (vs the previous idealized LJ fluid demonstrations) and compared to a direct driven-flow estimate. Extending these developments to treat nanoporous systems in order to directly quantify liquid–surface versus more bulk-like effects of confinement would benefit the understanding and design. As discussed in the previous section, the local environment of the ionic species in the IL has links to the overall transport efficiency as has been observed in pure ILs.^{13,47–50} Adapting the solvation persistence analysis of refs

13 and 50 to the nanoporous systems and correlating with the confinement effects could provide more definitive conclusions, regarding the link between vehicular versus structural transport and confinement. Furthermore, resolving boundary layers and variations in transport across slit pores can be revealing;^{4,41–43} however, corresponding methods for the irregular geometry of nanoporous systems have yet to be developed. Using pair-wise correlation of mean-squared displacement (or other transport measures) and radial distance from the matrix surface could yield a methodology analogous to the spatially binning used to resolve transport properties in regular geometries. Surface interactions certainly play a significant role in nanoporous transport, which suggests that surface charge state, determined by the mixture of silanol and siloxane terminations at the silica surface, may influence transport properties. Since this is a chemically realizable control of IG design with some promise to benefit performance, it also deserves future study. Similarly, it would be computationally feasible to explore the transport properties of mixed ILs, for example, combinations Pyr1.3 and Pyr1.4, under confinement. Disordering in mixed ILs is known to suppress glass transition temperatures,⁶ which may yield further relative enhancement of low-temperature transport coefficients in highly confined IGs. Last, estimating the electric field-driven ionic mobilities for the nanoporous systems would provide a counter-point to the equilibrium methods used in this study and may reveal how nanoporous morphologies affect Li⁺ counter currents observed in experimental and model systems.^{47–50}

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was funded by the Laboratory Directed Research and Development Program (project 218253). Sandia National Laboratories is a multimission laboratory managed and operated by National Technology and Engineering Solutions of Sandia, LLC, a wholly owned subsidiary of Honeywell International, Inc., for the U.S. Department of Energy's National Nuclear Security Administration under contract DE-NA-0003525. The views expressed in the article do not necessarily represent the views of the U.S. Department of Energy or the United States Government.

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